

ERDŐS PROBLEM 928: QUANTITATIVE COMPARISONS AND ORIGINAL FORMULATIONS

JI HO BAE

ABSTRACT. We prove explicitly that OpenAI's joint Dickman law implies the density formula $\rho(1/\alpha)\rho(1/\beta)$ in the precise formulation of Erdős Problem 928. Here ρ is the Dickman–de Bruijn function. For fixed $0 < \alpha, \beta < 1$ and real $X \geq 2$, an elementary argument bounds the symmetric difference between the counting sets in the problem and in the cited theorem, restricted to $2 \leq n \leq X$, by $\pi(X^\alpha) + \pi((X+1)^\beta)$, where π counts primes. This bound handles both the strict inequalities and the change in the second threshold from n^β to $(n+1)^\beta$. We also treat the strict formulation with both thresholds based at n , and compare the formulations in Erdős's two 1976 papers and the 1978 Erdős–Pomerance paper. The analytic joint law is due to OpenAI; our contribution consists of the explicit finite comparison estimates and the precise correspondence among these formulations.

1. INTRODUCTION

For an integer $n \geq 2$, let $P(n)$ be the largest prime factor of n , and set $P(1) = 1$. Write ρ for the Dickman–de Bruijn function: the continuous function on $[0, \infty)$ determined by

$$\rho(u) = 1 \quad (0 \leq u \leq 1), \quad u\rho'(u) = -\rho(u-1) \quad (u > 1).$$

The classical smooth-number limit is

$$(1) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{1 \leq n \leq X : P(n) \leq X^c\} = \rho(1/c) \quad (0 < c < 1);$$

see de Bruijn [2, p. 51, (1.1)–(1.2)]. We also use the standard properties $\rho(u) > 0$ for finite $u \geq 0$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$; see [2, p. 52, (1.8)] together with the defining equation. We use $\mathbb{N} = \{1, 2, \dots\}$. All counting variables are integers, and the endpoints X are real. Natural density means the limit of the ordinary, unweighted count up to X , divided by X .

Problem 928 in Bloom's collection [1] asks whether, for fixed $\alpha, \beta \in (0, 1)$, the integers satisfying

$$(2) \quad P(n) < n^\alpha, \quad P(n+1) < (n+1)^\beta$$

have a natural density. The accompanying discussion asks whether this density is $\rho(1/\alpha)\rho(1/\beta)$. The page cites Erdős [3, 4] and Erdős–Pomerance [5]; their formulations will be compared in Section 3.

Our analytic input is OpenAI's joint Dickman law [6, Theorem 1.1, p. 2]: for every fixed $\alpha, \beta \in (0, 1)$,

$$(3) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \leq n^\alpha, P(n+1) \leq n^\beta\} = \rho(1/\alpha)\rho(1/\beta).$$

OpenAI identifies its result as resolving the Erdős–Pomerance joint Dickman conjecture. Section 11.3 of that paper also proves the versions with thresholds X^α, X^β and the corresponding upper-tail conditions. We take these analytic results as given. Proposition 1 supplies explicit elementary estimates for the changes of convention needed in (2) and in Erdős's strict formulation. Section 3 identifies the precise original statements to which the results apply.

2. FINITE COMPARISON ESTIMATES

Fix $\alpha, \beta \in (0, 1)$. For $X \geq 2$, define

$$\begin{aligned}\mathcal{A}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < (n+1)^\beta\}, \\ \mathcal{B}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) \leq n^\alpha, P(n+1) \leq n^\beta\}, \\ \mathcal{C}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < n^\beta\}.\end{aligned}$$

The sets $\mathcal{A}$ and $\mathcal{C}$ correspond respectively to Problem 928 and condition (1) in [3, p. 25]; $\mathcal{B}$ is the set in (3). Let $\pi(y) = \#\{p \leq y : p \text{ is prime}\}$, and let $S \triangle T$ denote the symmetric difference of two sets.

Proposition 1 (Comparison of conventions). *For every $\alpha, \beta \in (0, 1)$ and every real $X \geq 2$,*

$$\begin{aligned}(4) \quad & \#(\mathcal{A}_{\alpha,\beta}(X) \triangle \mathcal{B}_{\alpha,\beta}(X)) \leq \pi(X^\alpha) + \pi((X+1)^\beta), \\ (5) \quad & 0 \leq \#\mathcal{B}_{\alpha,\beta}(X) - \#\mathcal{C}_{\alpha,\beta}(X) \leq \pi(X^\alpha) + \pi(X^\beta).\end{aligned}$$

Proof. Suppress the subscripts and the argument X . If $n \in \mathcal{B} \setminus \mathcal{A}$, then $P(n+1) \leq n^\beta < (n+1)^\beta$, so the second condition for $\mathcal{A}$ holds. The first condition must therefore fail, giving $P(n) = n^\alpha$. The prime $p = P(n) \leq X^\alpha$ determines $n = p^{1/\alpha}$ uniquely. Consequently $\#(\mathcal{B} \setminus \mathcal{A}) \leq \pi(X^\alpha)$.

If $n \in \mathcal{A} \setminus \mathcal{B}$, the first condition for $\mathcal{B}$ holds, so

$$n^\beta < P(n+1) < (n+1)^\beta.$$

The intervals $(m^\beta, (m+1)^\beta)$, for integers $m \geq 2$, are pairwise disjoint because $\beta > 0$. Thus $n \mapsto P(n+1)$ is injective on $\mathcal{A} \setminus \mathcal{B}$, and its values are primes less than $(X+1)^\beta$. This proves $\#(\mathcal{A} \setminus \mathcal{B}) \leq \pi((X+1)^\beta)$ and hence (4).

Finally, $\mathcal{C} \subseteq \mathcal{B}$. Any $n \in \mathcal{B} \setminus \mathcal{C}$ satisfies $P(n) = n^\alpha$ or $P(n+1) = n^\beta$. The first equality contributes at most $\pi(X^\alpha)$ integers, by the argument above. For the second, $n = P(n+1)^{1/\beta}$ is determined by a prime at most X^β , giving at most $\pi(X^\beta)$ integers. Summing these two bounds proves (5). $\square$

Corollary 2 (Problem 928 and Erdős's strict formulation). *For every fixed $\alpha, \beta \in (0, 1)$,*

$$(6) \quad \lim_{X \rightarrow \infty} \frac{\#\mathcal{A}_{\alpha,\beta}(X)}{X} = \lim_{X \rightarrow \infty} \frac{\#\mathcal{C}_{\alpha,\beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

In particular, the density in Problem 928 and the density $d(\alpha, \beta)$ asked for in [3, p. 25] both exist and equal the predicted product.

Proof. For finite sets, $|\#S - \#T| \leq \#(S \triangle T)$. Since $\pi(y) \leq \lfloor y \rfloor$, both upper bounds in Proposition 1 are $O(X^\alpha + X^\beta) = o(X)$. After division by X , equation (3) therefore gives both limits in (6). $\square$

The comparison estimates are independent of the joint Dickman law. They quantify the change of convention for fixed α, β ; the asymptotic evaluation of the count is supplied by (3).

3. COMPARISON WITH THE ORIGINAL SOURCES

We call a threshold *moving* when it depends on the integer being counted, and *fixed* when it is held constant over the counting range. Thus n^α and $(n+1)^\beta$ are moving thresholds, while X^α and X^β are fixed thresholds. Put

$$a(0) = 1, \quad a(t) = 1 - \rho(1/t) \quad (0 < t \leq 1).$$

Table 1 summarizes the two-integer formulations. The subsections below justify the changes of convention and counting range.

TABLE 1. The two-integer formulations and their deductions. Exponents are fixed: $0 < \alpha, \beta < 1$, $\gamma > 1$, and $0 \leq t, s \leq 1$. Unless indicated otherwise, $2 \leq n \leq X$.

Source	Conditions	Deduction
[1]; independence statement in [3, p. 25]	$P(n) < n^\alpha, P(n+1) < (n+1)^\beta$	Corollary 2, set $\mathcal{A}$: density $\rho(1/\alpha)\rho(1/\beta)$
[3, p. 25, (1)]	$P(n) < n^\alpha, P(n+1) < n^\beta$	Corollary 2, set $\mathcal{C}$: density $\rho(1/\alpha)\rho(1/\beta)$
[4, pp. 272–273], $r = 2$	$P(n+1) \leq k, P(n+2) \leq k; 1 \leq n < k^\gamma$	OpenAI [6, (11.7), p. 80]: count $\sim \rho(\gamma)^2 k^\gamma$
[5, p. 311]	$P(n) \geq X^t, P(n+1) \geq X^s$	OpenAI [6, (11.8) and following paragraph, p. 80]: count/ $X \rightarrow a(t)a(s)$

3.1. Erdős’s Manitoba paper. On page 25 of [3], Erdős expects independence for $P(n) < n^\alpha$ and $P(n+1) < (n+1)^\beta$. In the following sentence, he asks whether the integers satisfying his condition (1), namely $P(n) < n^\alpha$ and $P(n+1) < n^\beta$, have a density $d(\alpha, \beta)$. Thus the webpage uses the inequalities in his independence statement, whereas condition (1) uses n^β for the second threshold. Corollary 2 treats both forms explicitly.

Erdős also asks whether these strict inequalities have infinitely many solutions for each fixed pair $\alpha, \beta > 0$. The positive value in (6) implies this, also for parameters outside $(0, 1)$ by choosing smaller positive parameters inside that interval. Infinitude itself was already available: as recorded in [1], it follows from Schinzel [7, Theorem 14, p. 231], applied to $(2n+1)^2 - 1 = 4n(n+1)$. It gives a constant $K > 0$ and infinitely many integers $n \geq 3$ such that

$$P(n(n+1)) \leq n^{K/\log \log n}.$$

Since $K/\log \log n \rightarrow 0$, these integers satisfy Erdős’s strict inequalities for every prescribed positive pair α, β , once n is sufficiently large.

3.2. Erdős’s Debrecen paper. Write $\gamma > 1$ for the exponent denoted by α in [4, pp. 272–273]. Erdős recalls the one-variable asymptotic $(c_\gamma + o(1))k^\gamma$ for the number of $m \leq k^\gamma$ with $P(m) \leq k$, and conjectures, for each fixed integer $r \geq 1$, that as k tends to infinity through positive integers,

$$(7) \quad \#\{1 \leq n < k^\gamma : P((n+1) \cdots (n+r)) \leq k\} = (c_\gamma^r + o(1))k^\gamma.$$

Equation (1) identifies $c_\gamma = \rho(\gamma)$.

OpenAI [6, (11.7), p. 80] proves, for fixed $c, d \in (0, 1)$,

$$(8) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq m \leq X : P(m) \leq X^c, P(m+1) \leq X^d\} = \rho(1/c)\rho(1/d).$$

For $r = 2$ in (7), put $X = k^\gamma$, $m = n+1$, and $c = d = 1/\gamma$. Since $P(m(m+1)) = \max\{P(m), P(m+1)\}$, the prime-factor conditions coincide with those in (8). The ranges $1 \leq n < X$ and $2 \leq m \leq X$ differ, after the shift, by at most one endpoint term. Hence (8) gives (7) for $r = 2$.

The webpage [1] states the higher-order conjecture using $P(n+i) \leq n^\alpha$ for $0 \leq i < r$. Its case $r = 2$ is directly (3) with $\beta = \alpha$. The results used here concern two consecutive integers and do not establish the $r \geq 3$ assertion.

3.3. Erdős and Pomerance. On page 311 of [5], the authors use fixed thresholds and upper tails. For $0 \leq t, s \leq 1$, they ask whether

$$(9) \quad b(t, s) = \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \geq X^t, P(n+1) \geq X^s\}$$

exists, is continuous on $[0, 1]^2$, and equals $a(t)a(s)$, where their $a(t)$ is the corresponding one-variable upper-tail limit. Equation (1), together with the treatment of equality and endpoints below, identifies their $a(t)$ with the function defined above.

For $0 < t, s < 1$, inclusion–exclusion applied to (8) and the two marginal laws gives the limit $a(t)a(s)$ with strict upper tails $P(n) > X^t$, $P(n+1) > X^s$. For the shifted marginal, replacing n by $n+1$ changes the counting range by only $O(1)$ terms.

To pass to weak upper tails, observe that for fixed $0 < t \leq 1$ the equality $P(m) = X^t$ is impossible unless X^t is a prime p . In that case $p \mid m$, so there are at most $\lfloor (X+1)/p \rfloor = O(X^{1-t} + 1) = o(X)$ such $m \leq X+1$.

At exponent 0 the upper-tail condition is automatic for $m \geq 2$. At exponent 1, within the range $2 \leq m \leq X+1$, the condition $P(m) \geq X$ forces $m \in [X, X+1]$, so at most two integers occur. This proves (9) at all endpoints as well. Since $\rho(1) = 1$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$, the function a is continuous on $[0, 1]$, and so is $b(t, s) = a(t)a(s)$.

This deduction, including the treatment of equality and endpoints, is already given in [6, (11.8) and following paragraph, p. 80]. The role of the present comparison is to match it to the exact statement in [5]. All the two-integer formulations in Table 1 therefore follow from the cited joint Dickman law and its fixed-threshold version.

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